

Abiodun Victor Alagbada

PhD Candidate in Applied Mathematics — Researcher — Educator — Developer

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Profile

PhD candidate in Applied Mathematics at Bauhaus University Weimar, working on Koopman-operator learning and hybrid physics–data frameworks for structural health monitoring (SHM). Our research develops interpretable, data-driven dynamical system models for vibration-based condition assessment and damage diagnosis, with an emphasis on system identification, time-series learning, and reduced-order representations. We build reproducible computational workflows in Python for scalable experimentation, model evaluation, and research-to-deployment translation. Experienced educator and developer with a strong record of curriculum delivery, student mentorship, and the development of web-enabled research tooling.

Professional Experience

Full-Stack Developer / Backend Engineer

2022 – Present

- Designed and deployed scalable web and AI-based academic tools using Python (Django, Flask) and JavaScript (React, Node.js).
- Implemented RESTful APIs and containerized pipelines with Docker and GitHub Actions.
- Contributed to computational frameworks supporting research data automation and model testing.

Mathematics and Computer Science Teacher — Omega High School, Nigeria

2018 – 2023

- Taught Mathematics, Advanced Mathematics, and Computer Science.
- Integrated technology, programming, and data visualization tools into teaching practice.
- Led assessment moderation, curriculum planning, and mentoring initiatives.

Online Mathematics Tutor — Preply and Upwork

2020 – Present

- Delivered personalized lessons and academic coaching to learners across diverse backgrounds.
- Developed digital teaching workflows using Jupyter, LaTeX, and computational visualization tools.

Leadership and Soft Skills

- Proven leadership experience (financial secretary, academic project lead).
- Excellent communication, teamwork, and instructional skills.
- Strong analytical reasoning, attention to detail, and problem-solving abilities.

Education

PhD in Applied Mathematics (Ongoing) — Bauhaus University Weimar, Germany
2024 – Present

Focus: Koopman Analysis, Structural Health Monitoring, Data-Driven Simulation, Computational Methods, and Optimization.

Software Engineering (Backend Development) — ALX Online Program
2023 – 2025

MSc in Mathematics — University of Jos, Nigeria
2017 – 2018

BSc in Mathematics — University of Jos, Nigeria
2010 – 2014, Second Class Upper

Further Education and Certification

- Qualified Teacher Status (QTS) — UK Department for Education (TRN: 2361519)
- Web Design Consultancy — Nova University (2023)
- Backend Web Developer Course — Side Hustle (2022)
- Full-Stack Developer Course — Opecstat Academy (2022)

Research Experience and Publications

PhD Research — Bauhaus University Weimar

Koopman-operator learning and hybrid physics–data frameworks for structural health monitoring (SHM), focusing on interpretable dynamical representations for vibration-based condition assessment and damage diagnosis. We develop system identification and time-series learning pipelines, emphasizing reproducibility, benchmarking, and scalable experimentation in Python.

Publications and Manuscripts:

- **Book Chapter (Approved):** “Machine Learning Theoretical Foundations, Practical Applications, Ethical Implications, And Future Prospects: Exploring Challenges And Applications In Emerging Technologies (221124-055235).” In: *Encyclopedia of Modern Artificial Intelligence*.

- **Manuscript (Preprints):** “Neural Regime-Switching Model with Markov-Informed Attention for Short-Horizon Equity Return Forecasting.”
- **Manuscript (Approved):** “Koopman Operator Methods in Structural Health Monitoring: A Systematic Review Towards Hybrid Physics–Data Frameworks.” (Manuscript ID: applsci-4214533).

MSc Thesis — University of Jos (2018)

A Class of Block Second Derivative Method for Solving Stiff Ordinary Differential Equations.

Supervisor: Prof. Kumleng Geoffrey.

BSc Thesis — University of Jos (2014)

The Convergence of Some Complex Power Series.

Supervisor: Dr. Silas Longwap.

Technical Skills

Programming Languages	Python (Advanced), C/C++ (Intermediate), JavaScript (Intermediate), SQL (Intermediate), MATLAB (Intermediate)
Frameworks	Django (Intermediate), Flask (Intermediate), React (Intermediate), Node.js (Intermediate)
Scientific Computing	NumPy/Pandas (Advanced), TensorFlow (Intermediate), Scikit-learn (Intermediate), Jupyter/Marimo (Advanced)
Tools	Docker (Intermediate), Git/GitHub (Advanced), LaTeX (Advanced)

Languages

English (Native) · German (A2–B1) · Yoruba (Native)

Professional Memberships

- Mathematics Association of Nigeria
- Black in Mathematics, South Africa
- American Mathematical Society (AMS)

Honors and Awards

- PTDF Scholarship (Nigeria) — 2023/2024 Cohort
- DAAD Scholarship — 2025
- Bronze Medal – National Mathematics Competition for University Students (2013)
- Certificate of Participation – National Mathematics Competition (2013)

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Weimar, Germany

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